Sahaj Bhadja

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Summary

Early-career Software Engineer graduating May 2026 with 2+ years building **distributed systems**, concurrent data pipelines, and cross-platform services using **C++, Python, Linux, AWS, Azure**, CI/CD, and production testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Architected multi-tenant data infrastructure on Azure SQL and Blob Storage with row-level isolation, supporting 500+ users while reducing production indexed-query latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous orchestration with explicit state transitions and failure isolation, processing 1K+ jobs while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Designed a concurrent data-processing pipeline using multithreading and secure file ingestion, processing 1K+ files per batch through AWS RDS and REST APIs with isolated, observable, fault-tolerant execution.
- Implemented production Python services handling 10K+ daily requests for 150+ users, using CI/CD-backed workflows to reduce turnaround 60% while preserving stable application performance during peak production load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*
- Built a Python autonomous-driving simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% while rigorously testing data transformations and model behavior across repeatable vehicle benchmark scenarios.
- Architected a spatial-temporal encoder combining map geometry and vehicle trajectory context, improving closed-loop stability 15% through systematic simulation, regression testing, detailed production failure analysis, root-cause investigation, and debugging.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Engineered cross-platform microservices with automated VM builds and 20+ robust unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments.
- Optimized object-oriented API integrations and AWS Glue and S3 data flows, improving response times 30%, pipeline efficiency 68%, and long-term maintainability through reusable Factory and Singleton patterns.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellow *India*
- Evaluated distributed consensus algorithms and designed a quantum-resistant protocol integrating five lattice-based cryptographic primitives, reducing centralized-control and simulated quantum-risk exposure by 25% across repeatable distributed benchmark workloads.
- Validated protocol behavior through repeatable benchmark simulations, documented findings in a peer-reviewed IEEE technical paper, and formally presented the distributed-systems research, methodology, and results at IISc Bengaluru.

Skills

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL, Bash
Systems: Linux, Distributed Systems, Configuration Management, Data Pipelines, Concurrency, Multithreading, OOP, Operating Systems, Algorithms, Data Structures, REST APIs
Cloud & Data: AWS, Azure, S3, Blob Storage, PostgreSQL, SQL Server, MySQL, Docker, Kubernetes, Cloud Storage, ETL
Quality: Git, CI/CD, Jenkins, GitHub Actions, PyTest, Mockito, Unit Testing, Integration Testing, Regression Testing, Observability, Debugging, Agile

Projects

FleetRelay | Python, Embedded Data Plane, Configuration, Telemetry
- Built and tested a resource-bounded vehicle agent with offline buffering, reconnect-aware batch upload, certificate-derived identity, signed versioned configuration, payload limits, and telemetry counters for failures and delivery.

CPU-scheduling Simulator Live | C++, React.js, Operating Systems
- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling algorithms with interactive visualization and dynamic waiting-time and turnaround-time calculations across 1K+ repeatable simulated process combinations and edge cases.

Linguistix | Python, AWS, Docker, CI/CD
- Developed and deployed a containerized data-processing API using AWS CodePipeline and ECS Fargate, adding automated releases and validation checks while achieving 87% processing accuracy and production-grade reliability.